

INFORMATION SYSTEM ANALYSIS AND DESIGN

ASSIGNMENT 01

E-Commerce Information System Investigation

Course	Information System Analysis and Design
Assignment	Assignment 01
Topic	E-Commerce Information System Investigation
Duration	2 weeks
Suggested workload	6–8 student hours
Working mode	Individual
Submission	PDF report
Recommended length	8-10 pages, excluding references and appendices

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1 Assignment Overview

Assignment 01 introduces students to the first and most important stage of Information System Analysis and Design: **understanding the real-world problem before designing the software system.**

In this assignment, students investigate a realistic e-commerce business and analyze its:

- business context;
- stakeholders;
- major business activities;
- information;
- current problems;
- system scope;
- system boundary;
- preliminary information-system objectives.

The assignment deliberately does not require detailed UML modeling or implementation. Students should first understand *why* an information system is needed and *what business problem* it should address.

Central question of Assignment 01

What problem should the information system solve?

The work produced in this assignment will become the foundation for Assignment 02, where students transform the identified problems and stakeholder needs into formal system requirements.

2 Learning Objectives

After completing this assignment, students should be able to:

1. Describe an organization's business context and information-system needs.
2. Identify and classify stakeholders involved in an e-commerce system.

3. Analyze major business activities and information flows.
4. Identify important information produced, consumed, and stored by the organization.
5. Identify current problems, their causes, and their consequences.
6. Define the preliminary scope and boundary of an information system.
7. Formulate clear objectives for a proposed information system.
8. Present an information-system investigation in a structured academic report.

Expected learning progression

Organization → Business Activities → Stakeholders → Information → Problems → System Scope

3 Case Study: ABC Online Shop

3.1 Business Background

ABC Online Shop is a medium-sized business selling consumer products through online channels.

The company manages several major business activities, including:

- product management;
- product category management;
- customer management;
- shopping-cart management;
- order processing;
- payment processing;
- inventory management;
- shipping and delivery;
- customer reviews;
- promotions;

- sales reporting.

The business interacts with customers, employees, suppliers, payment providers, and shipping providers.

3.2 Current Problems

The company currently experiences several information-management problems:

- product information is maintained inconsistently;
- inventory information is sometimes outdated;
- employees manually process some orders;
- customers cannot always obtain accurate order-status information;
- sales reports require manual preparation;
- customer information is stored in different places;
- management has limited access to real-time business information.

These problems provide the initial motivation for investigating a new **E-Commerce Information System**.

Students should not simply repeat the case description.

They are expected to analyze the business situation, identify relationships among problems, determine who is affected, and explain why an information system may be necessary.

4 Student Task

Students must produce an **E-Commerce Information System Investigation Report** for ABC Online Shop.

The report should answer the following six questions:

1. What is the business?
2. Who participates in the business?
3. What does the business do?
4. What information does the business use?

5. What problems exist?
6. What system should be considered?

The analysis should move from the real-world business context toward a preliminary information-system proposal.

5 Part I — Business Background

Describe the organization and its business model.

Your analysis should include:

- organization name;
- type of business;
- products or services;
- target customers;
- sales channels;
- major business activities;
- business objectives.

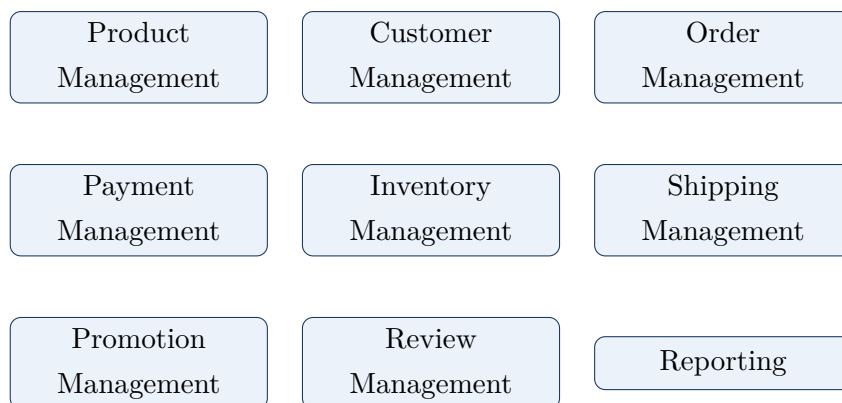
Students should answer:

What is the primary business purpose of the organization?

The answer should be expressed in one or two clear paragraphs.

5.1 Major Business Activities

At a high level, the business may include:



Students should explain the role of each major activity.

6 Part II — Stakeholder Analysis

Identify all important stakeholders who interact with or are affected by the e-commerce business.

Students should distinguish between:

6.1 Internal Stakeholders

Examples include:

- administrator;
- sales staff;
- warehouse staff;
- manager.

6.2 External Stakeholders

Examples include:

- customers;
- suppliers;
- payment providers;
- shipping providers.

6.3 Stakeholder Table

Complete a table similar to Table 1.

Table 1: Stakeholder analysis

Stakeholder	Role	Main Interest / Need
Customer	Purchases products	Convenient and reliable online shopping
Administrator	Manages business/system	Efficient and accurate management
Sales Staff	Handles orders/customers	Accurate customer and order information
Warehouse Staff	Manages inventory	Accurate stock information
Manager	Monitors business	Reports and decision-support information
Payment Provider	Processes payments	Secure transaction information
Shipping Provider	Delivers products	Correct delivery information
Supplier	Provides products	Product and supply information

Students may add additional stakeholders where appropriate.

7 Part III — Business Activity Analysis

Identify and describe the major business activities of the e-commerce organization.

At minimum, analyze:

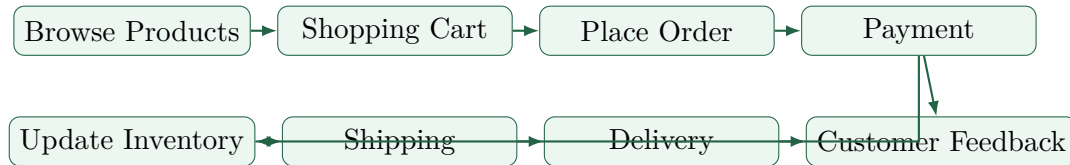
1. Product Management
2. Customer Management
3. Product Browsing
4. Shopping Cart
5. Order Processing
6. Payment
7. Inventory Management

8. Shipping

9. Customer Feedback

10. Reporting

A simplified business flow is:



7.1 Example: Order Processing

A high-level order process may be:

1. Customer places an order.
2. Order information is received.
3. Product availability is checked.
4. Payment is processed.
5. Order is prepared.
6. Inventory is updated.
7. Order is sent to shipping.
8. Order status is updated.
9. Customer receives confirmation.

Students should analyze the process rather than simply copy it.

8 Part IV — Information Analysis

Information is the central resource of an information system.

Students must identify information that is:

- produced;
- consumed;
- stored;

- exchanged.

8.1 Examples

Information produced:

- order confirmation;
- invoice;
- sales report;
- inventory report;
- shipping notification.

Information consumed:

- product information;
- customer information;
- inventory information;
- payment information;
- shipping information.

Information stored:

- customer information;
- product information;
- order information;
- payment information;
- inventory information;
- shipment information;
- review information.

8.2 Information Catalogue

Students should create an information catalogue similar to Table 2.

Table 2: Example information catalogue

Information	Produced By	Used By	Purpose
Product Information	Administrator	Customer	Product browsing
Customer Information	Customer	Staff	Order processing
Order Information	Customer/System	Staff	Order fulfillment
Inventory Information	Warehouse	Staff/System	Stock management
Payment Information	Payment Service	Order System	Payment processing
Sales Information	System	Manager	Decision support

Students should add additional information items where appropriate.

9 Part V — Current Problems

Students must identify at least **five significant problems** in the current business situation.

Do not merely list symptoms.

For each problem, analyze:

Problem → Cause → Consequence

For example:

Table 3: Example problem analysis

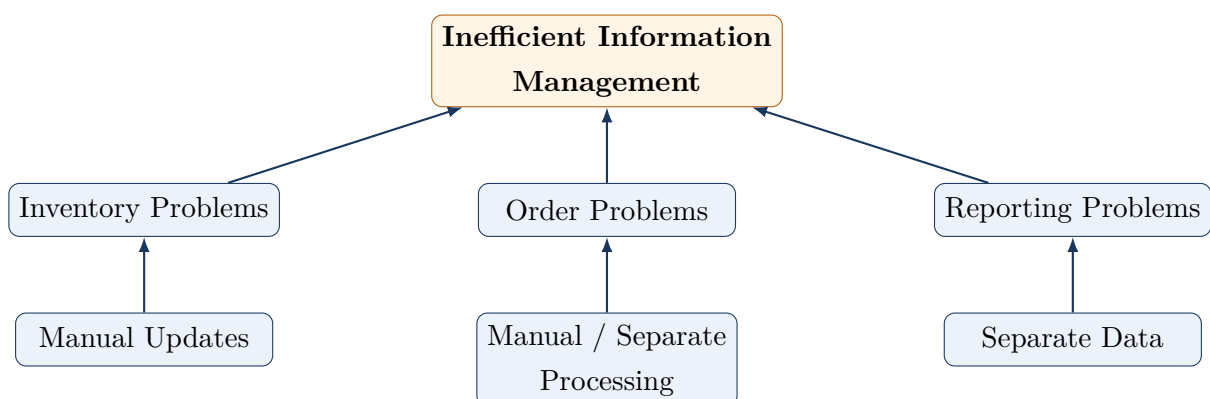
Problem	Possible Cause	Consequence
Inventory is inaccurate	Manual updates	Customers may order unavailable products
Order status is unclear	Information stored separately	Customers frequently contact staff
Reports are slow	Manual data collection	Management decisions are delayed
Duplicate customer data	Multiple files/systems	Inconsistent information
Product information is inconsistent	Different staff maintain data	Poor customer experience

Students should improve or replace these examples based on their own analysis.

9.1 Problem Tree

Students must create a problem tree showing relationships among major problems, causes, and consequences.

A simplified example is shown below.



Students should create a more complete problem tree based on their own investigation.

10 Part VI — Proposed Information System

Based on the investigation, students should propose a preliminary **E-Commerce Information System**.

The proposed system may support:

- customer management;
- product management;
- category management;
- shopping cart;
- order management;
- payment management;
- inventory management;
- shipping management;
- review management;
- promotion management;
- reporting.

At this stage, students should not design the internal implementation.

The objective is to determine the **scope of the system**.

11 Part VII — System Scope and Boundary

Students must define what belongs inside the proposed information system and what remains outside the system.

11.1 Inside the System

Possible functions include:

- customer management;
- product management;
- shopping cart;
- order management;
- inventory management;
- reporting.

11.2 External Systems

Possible external systems include:

- payment gateway;
- shipping provider;
- supplier system;
- external notification service.

Students must justify their boundary decisions.

11.3 Preliminary Context Diagram

Students must create a context diagram showing the proposed information system and its major external entities.

A simplified example is shown in Figure 1.

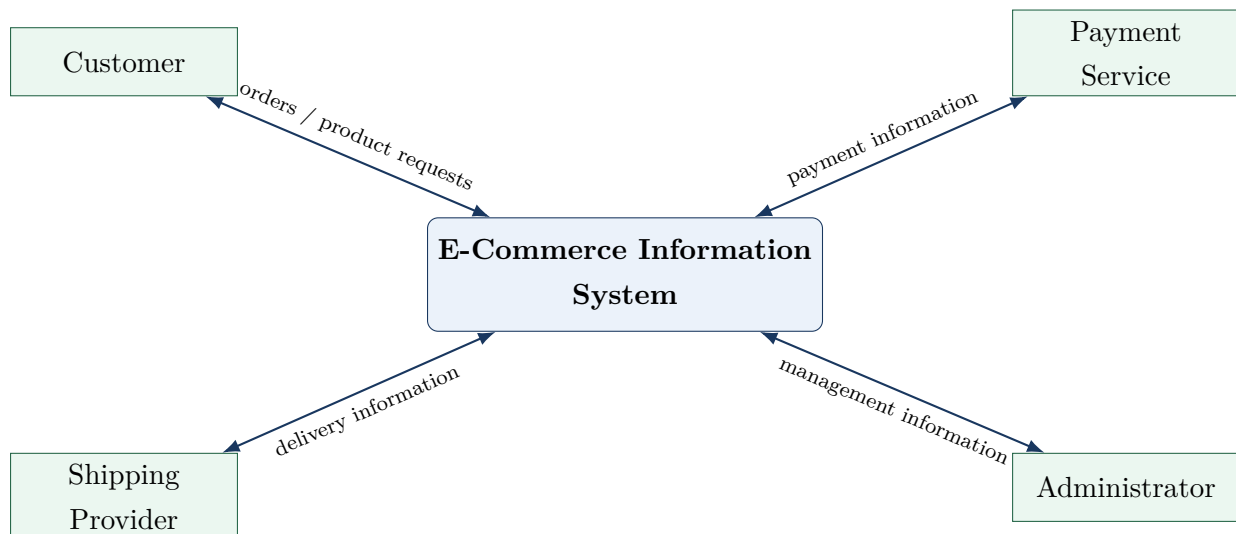


Figure 1: Preliminary context diagram

Students must create their own improved version and clearly label the major information flows.

12 Information Flows

Students should identify important information exchanged between the system and external entities.

For example:

Customer → System

- registration information;
- login information;
- product search;
- shopping-cart information;
- order information;
- review information.

System → Customer

- product information;
- order confirmation;
- payment status;
- order status;
- shipping information.

System → Payment Service

- payment request;
- transaction information.

Payment Service → System

- payment result;
- transaction status.

Students should identify additional flows where appropriate.

13 Part VIII — System Objectives

Students must formulate **5–7 system objectives**.

The objectives should be clear and connected to the problems identified earlier.

Examples include:

1. Provide customers with online product browsing and searching.
2. Support electronic shopping-cart and order-processing activities.
3. Maintain consistent product and inventory information.
4. Provide customers with order-status information.
5. Support secure integration with external payment services.
6. Provide management with sales and inventory reports.
7. Reduce manual processing of routine e-commerce activities.

Students should not simply copy these objectives. They should adapt them to their analyzed business situation.

14 Required Diagrams

Assignment 01 requires exactly three core diagrams.

14.1 Diagram 1 — Stakeholder Map

The diagram should show the main internal and external stakeholders.

14.2 Diagram 2 — High-Level Business Process

The diagram should show the main business activities and their relationships.

14.3 Diagram 3 — System Context Diagram

The diagram should show:

- the proposed information system;
- external entities;
- major information flows;
- the system boundary.

Important

Do not create a detailed UML Use Case Diagram in Assignment 01.

Use Case modeling will be introduced systematically in **Assignment 02 — Requirements Engineering**.

15 Required Report Structure

Students should use the following structure.

1. Introduction
2. Business Background
 - 2.1. Organization
 - 2.2. Business Model
 - 2.3. Products and Customers
 - 2.4. Major Business Activities
3. Stakeholder Analysis
 - 3.1. Internal Stakeholders
 - 3.2. External Stakeholders
 - 3.3. Stakeholder Interests
4. Business Activity Analysis
 - 4.1. Product Management
 - 4.2. Customer Management
 - 4.3. Order Processing
 - 4.4. Payment
 - 4.5. Inventory
 - 4.6. Shipping
 - 4.7. Reporting
5. Information Analysis

- 5.1. Information Produced
- 5.2. Information Consumed
- 5.3. Information Stored
- 5.4. Information Catalogue
- 6. Current Problems
 - 6.1. Problem Identification
 - 6.2. Problem Causes
 - 6.3. Problem Consequences
 - 6.4. Problem Tree
- 7. Proposed Information System
 - 7.1. System Purpose
 - 7.2. System Scope
 - 7.3. System Boundary
 - 7.4. Context Diagram
 - 7.5. Information Flows
 - 7.6. System Objectives
- 8. Conclusion
- 9. References
- 10. Appendices, if necessary

16 Submission Requirements

16.1 Main Report

Submit a PDF report of approximately:

6–8 pages

excluding references and appendices.

The report should be:

- logically structured;
- clearly written;
- professionally formatted;
- supported by appropriate diagrams;
- consistent in terminology.

16.2 Diagrams

All diagrams must:

- be readable;
- have a title or caption;
- be numbered;
- be referenced in the report;
- use consistent terminology.

For example:

Figure 3 presents the preliminary context of the proposed e-commerce information system.

16.3 File Naming

Use:

A01_StudentID_Name.pdf

17 Assessment Rubric

Assignment 01 is assessed on a 100-point scale.

Table 4: Assignment 01 assessment rubric

Criterion	Points	Expected Evidence
Business/domain understanding	15	Clear understanding of the e-commerce business and its objectives.
Stakeholder analysis	15	Relevant internal and external stakeholders with roles and interests.
Business activity analysis	15	Major business activities and their relationships are correctly identified.
Information analysis	15	Important information produced, consumed, stored, and exchanged is identified.
Problem analysis	20	Problems, causes, and consequences are logically analyzed.
System boundary and context	10	Clear system scope, boundary, external entities, and information flows.
System objectives	5	Clear objectives connected to identified problems.
Report quality and presentation	5	Clear writing, organization, diagrams, formatting, and consistency.
Total	100	

18 Performance Levels

Table 5: General performance levels

Score	Description
90–100	Excellent: comprehensive, logically connected, realistic, and well-presented analysis.
75–89	Good: mostly complete analysis with good logical consistency and minor gaps.
60–74	Satisfactory: basic understanding but with noticeable gaps or weaknesses.
40–59	Weak: incomplete, superficial, or partially incorrect analysis.
Below 40	Poor: major misunderstanding, missing analysis, or insufficient evidence.

19 E-Commerce Domain Variation

To encourage independent analysis, students may work with different e-commerce specializations while following the same analysis methodology.

Possible domains include:

Table 6: Possible e-commerce specializations

No.	Domain	Possible Special Characteristics
1	Fashion E-Commerce	Size, color, product variants
2	Electronics E-Commerce	Specifications, warranty, serial numbers
3	Book E-Commerce	Authors, publishers, editions, ISBN
4	Grocery E-Commerce	Weight, expiry date, stock availability
5	Furniture E-Commerce	Dimensions, materials, delivery requirements
6	Sports Equipment	Product specifications, sizes, brands
7	Cosmetics	Product variants, ingredients, categories
8	Multi-Vendor Marketplace	Multiple sellers, commissions, seller management

The purpose of this variation is to ensure that students apply the **same information-system methodology to different business contexts**.

20 Academic Guidance

20.1 Analysis Before Design

Students should not begin by designing database tables or writing program code.

The recommended reasoning process is:

Understand the Business → Identify Problems → Identify Stakeholders → Understand Information →

Only after this analysis should students proceed to requirements engineering.

20.2 Avoid Solution-First Thinking

A common mistake is:

“I will build an e-commerce website.”

Instead, students should reason:

Business Problem → Information Need → System Need → Potential Solution

20.3 Maintain Terminology Consistency

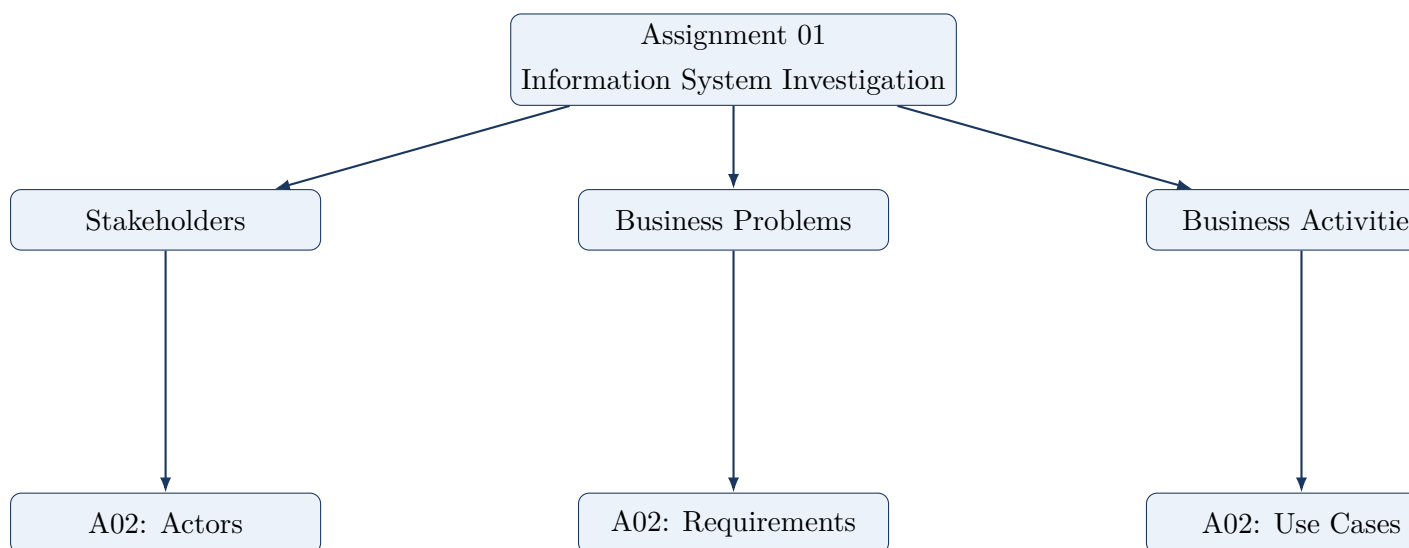
If a student calls something “Customer” in one section, the same concept should not suddenly become “Buyer” or “User” without explanation.

Consistent terminology will become increasingly important in Assignments 02–05.

21 Connection to Assignment 02

Assignment 01 is the foundation for Assignment 02.

The outputs should be reused as follows:



The course therefore follows a traceable development of knowledge:

Business Evidence → Requirements → Analysis Models → Design Models

22 Student Submission Checklist

Before submitting, students should verify the following.

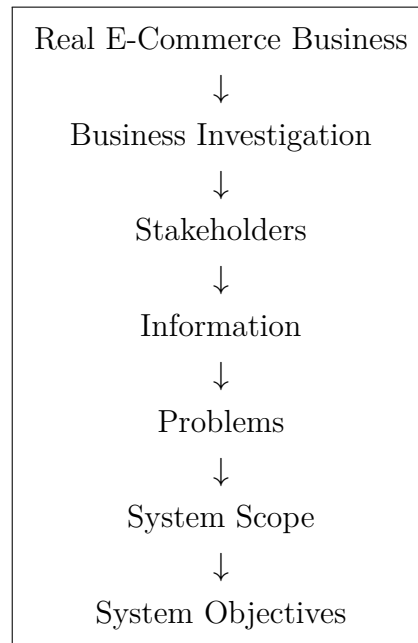
Table 7: Assignment 01 checklist

Done	Requirement
☐	Business background is clearly described.
☐	Major business activities are identified.
☐	Internal stakeholders are identified.
☐	External stakeholders are identified.
☐	Stakeholder interests are explained.
☐	Important information is identified.
☐	Information catalogue is provided.
☐	At least five significant problems are analyzed.
☐	Problem causes and consequences are explained.
☐	Problem tree is provided.
☐	Proposed system scope is defined.
☐	System boundary is explained.
☐	Stakeholder map is provided.
☐	High-level business process is provided.
☐	Context diagram is provided.
☐	Information flows are identified.
☐	Five to seven system objectives are formulated.
☐	Report is 6–8 pages approximately.
☐	All diagrams are numbered and referenced.
☐	PDF file follows the required naming convention.

23 Conclusion

Assignment 01 establishes the foundation for the complete Information System Analysis and Design process.

The expected progression is:



In Assignment 02, students will continue from these results and transform business needs into a formal set of **functional and non-functional requirements**, together with the initial **Use Case Model**.

Do not design the software before understanding the problem.

First understand the business. Then define the requirements.